K.R. MANGALAM UNIVERSITY

THE COMPLETE WORLD OF EDUCATION

LINUX BASH SCRIPTING LAB

(ENCS356)

Lab file Submitted to

K.R. MANGALAM UNIVERSITY

For

Bachelor Of Technology

In

Computer Science Engineering Specialization

(Cyber Security) Submitted By:

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Experiment No. 1

Problem Statement

Linux is a foundational platform for system administration, cloud computing, and cyber security applications. However, directly installing Linux on physical hardware may not always be feasible in an academic laboratory environment due to hardware constraints and system dependency issues. To address this challenge, virtualization technologies such as Oracle VirtualBox and VMware Workstation allow the deployment of Linux in an isolated and controlled virtual environment.

The problem is to install and configure a Linux operating system using a virtual machine manager (VirtualBox or VMware Workstation) with an OVA file, ensuring that the system is correctly set up and ready for performing Linux command-line and Bash scripting experiments.

1. Aim:

To install and configure a Linux operating system using a virtualization tool (Oracle VirtualBox or VMware Workstation) by importing an OVA file and preparing the system for Linux command-line and Bash scripting experiments.

2. Objective:

- To understand virtualization concepts
- To import and deploy a Linux OVA appliance
- To configure VM settings (RAM, CPU, Network)
- To verify Linux installation
- To prepare the environment for command-line and Bash scripting tasks

3. Software Requirements:

- Host System: Windows / macOS / Linux
- Virtualization Software:
    - Oracle VirtualBox

- VMware Workstation

- Linux OVA File

4. Theory:

4.1 Virtualization:

Virtualization allows multiple operating systems to run on a single physical machine using virtual hardware. A Virtual Machine (VM) behaves like a real computer but runs inside software.

4.2 OVA File:

OVA (Open Virtual Appliance) is a pre-configured virtual machine package that includes:

- Operating system

- Pre-installed tools

- Configurations

Importing an OVA saves time compared to manual installation.

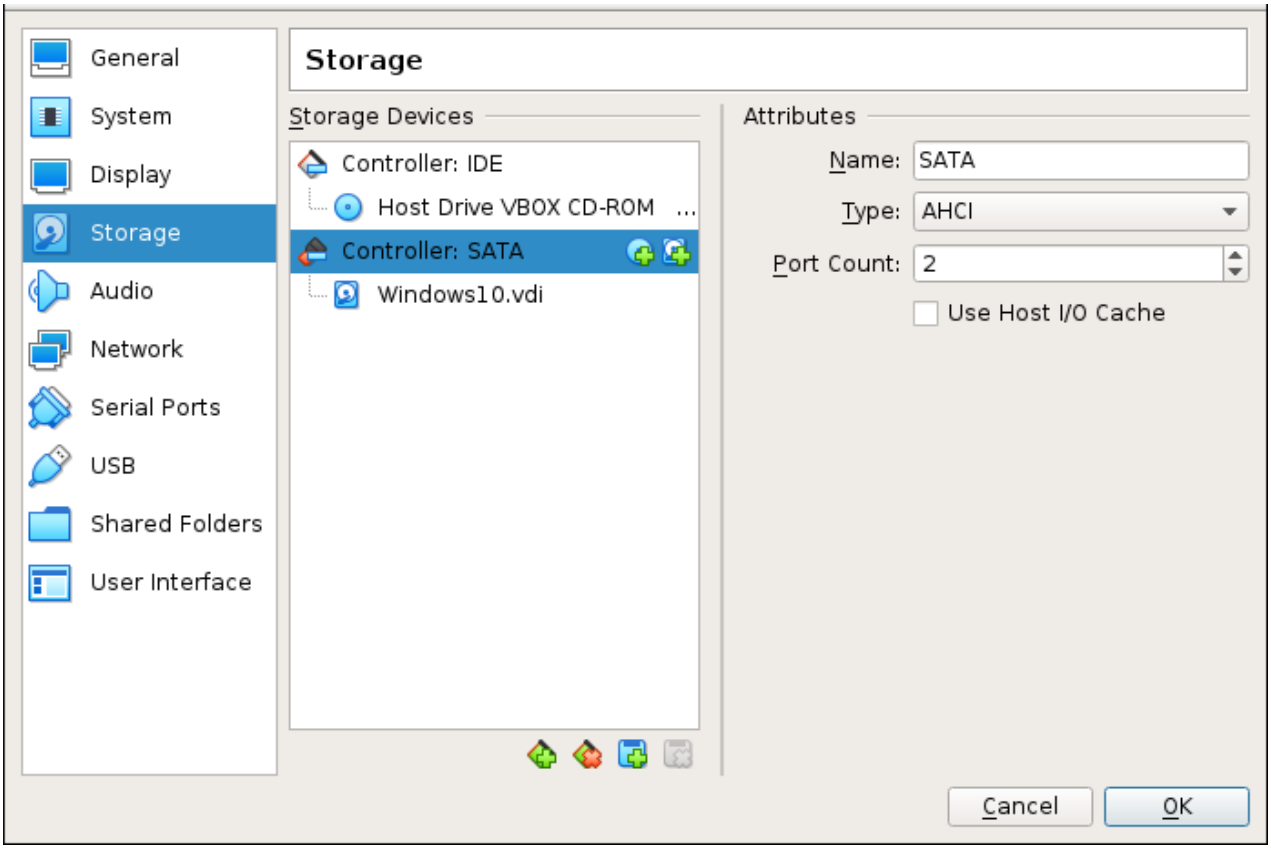
5. Procedure (Using VirtualBox):

PART A – Installation Using Oracle VirtualBox

Step 1: Install VirtualBox

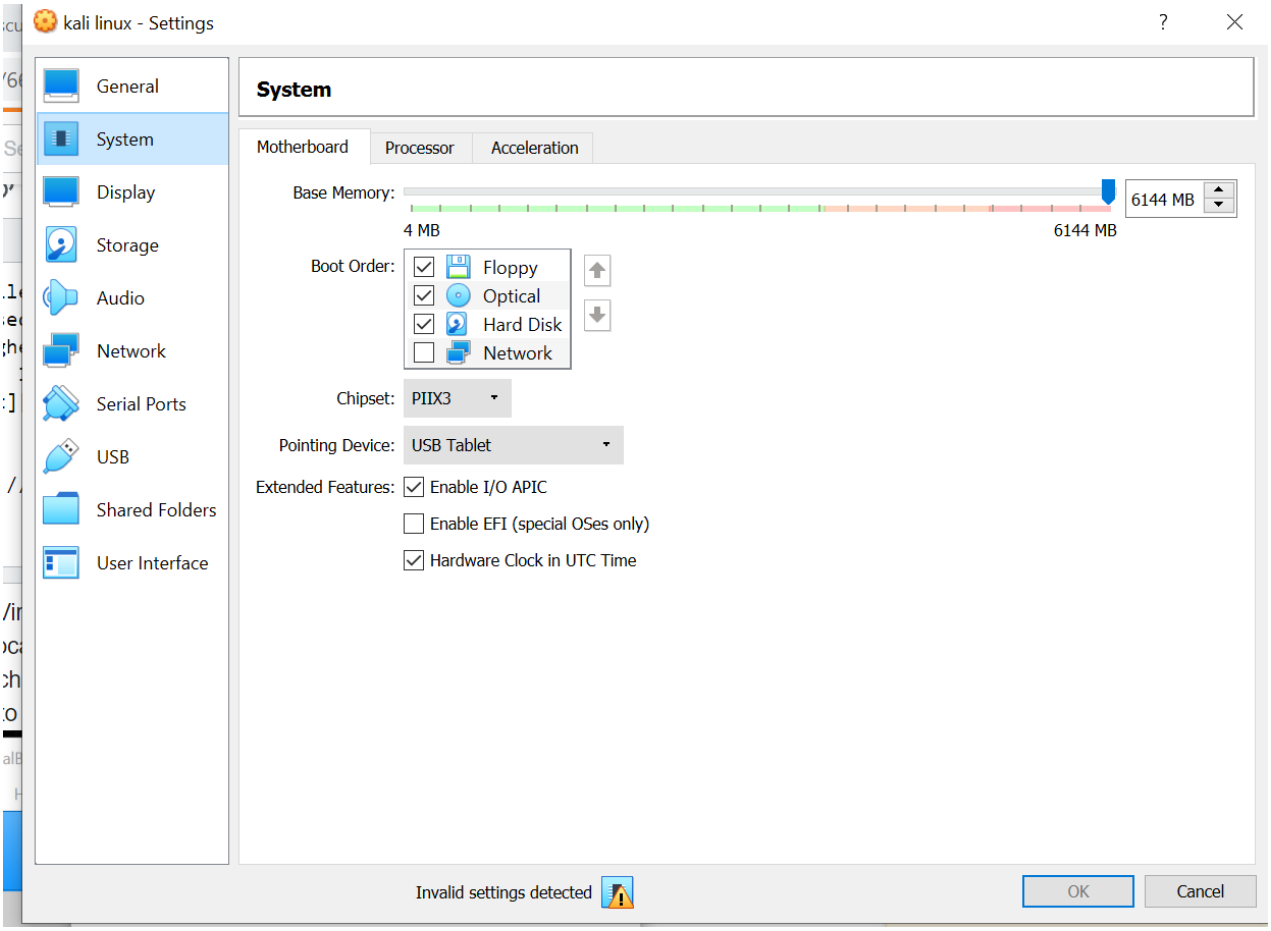
- Download VirtualBox from official website.
- Install using default settings.
- Launch VirtualBox Manager

Step 2: Import the OVA File



- Click **File → Import Appliance**.
- Select the provided .ova file.
- Click **Next**.
- Review appliance settings.
- Click **Import**.
- Wait until the import process completes.

Step 3: Modify VM Settings

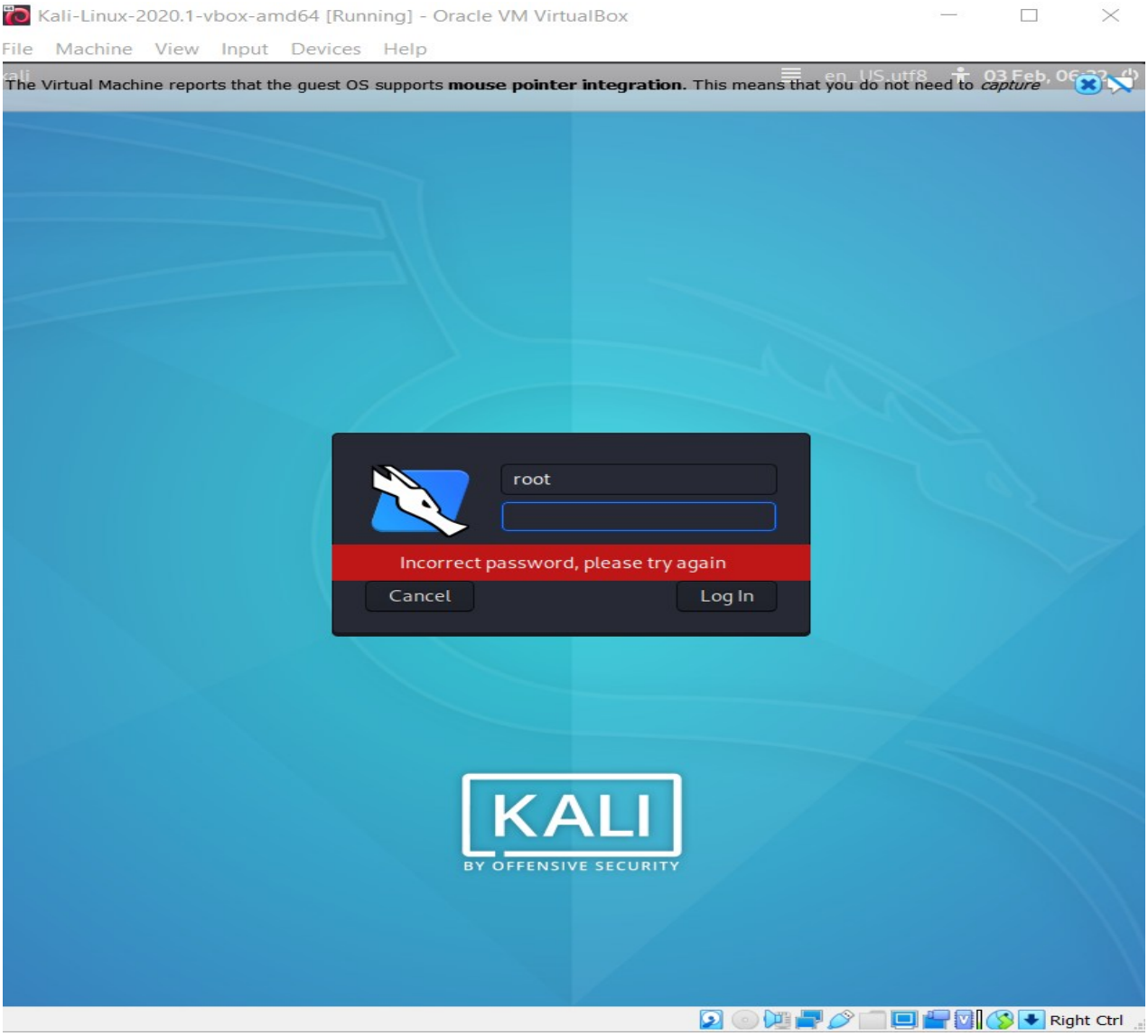


1. Select the VM → Click **Settings**.
2. Under **System**:
   - Set RAM to at least 2048 MB.
   - Assign 2 CPU cores (if available).
3. Under **Network**:

   - Set Adapter to **NAT** (recommended).

4. Click **OK**.

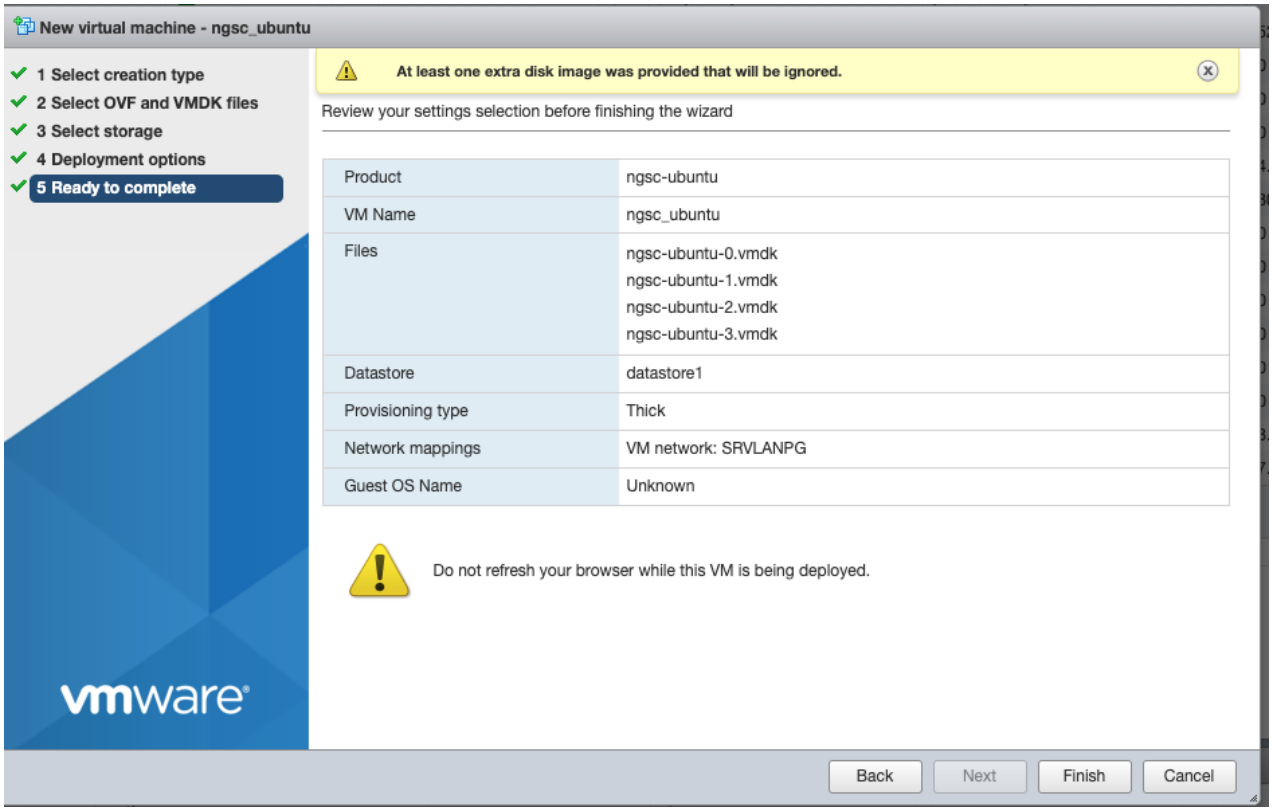
Step 4: Start the Virtual Machine



1. Click **Start**.
2. Wait for Linux to boot.
3. Log in using provided credentials.
4. Open **Terminal**.

Procedure (Using VMware Workstation):

Step 1: Open VMware



1. Open VMware Workstation.
2. Click Open a Virtual Machine.
3. Select the .ova file.
4. Follow import wizard.
5. Click Finish.

Step 2: Configure VM Settings:

1. Select VM → Click Edit virtual machine settings.

2. Allocate:

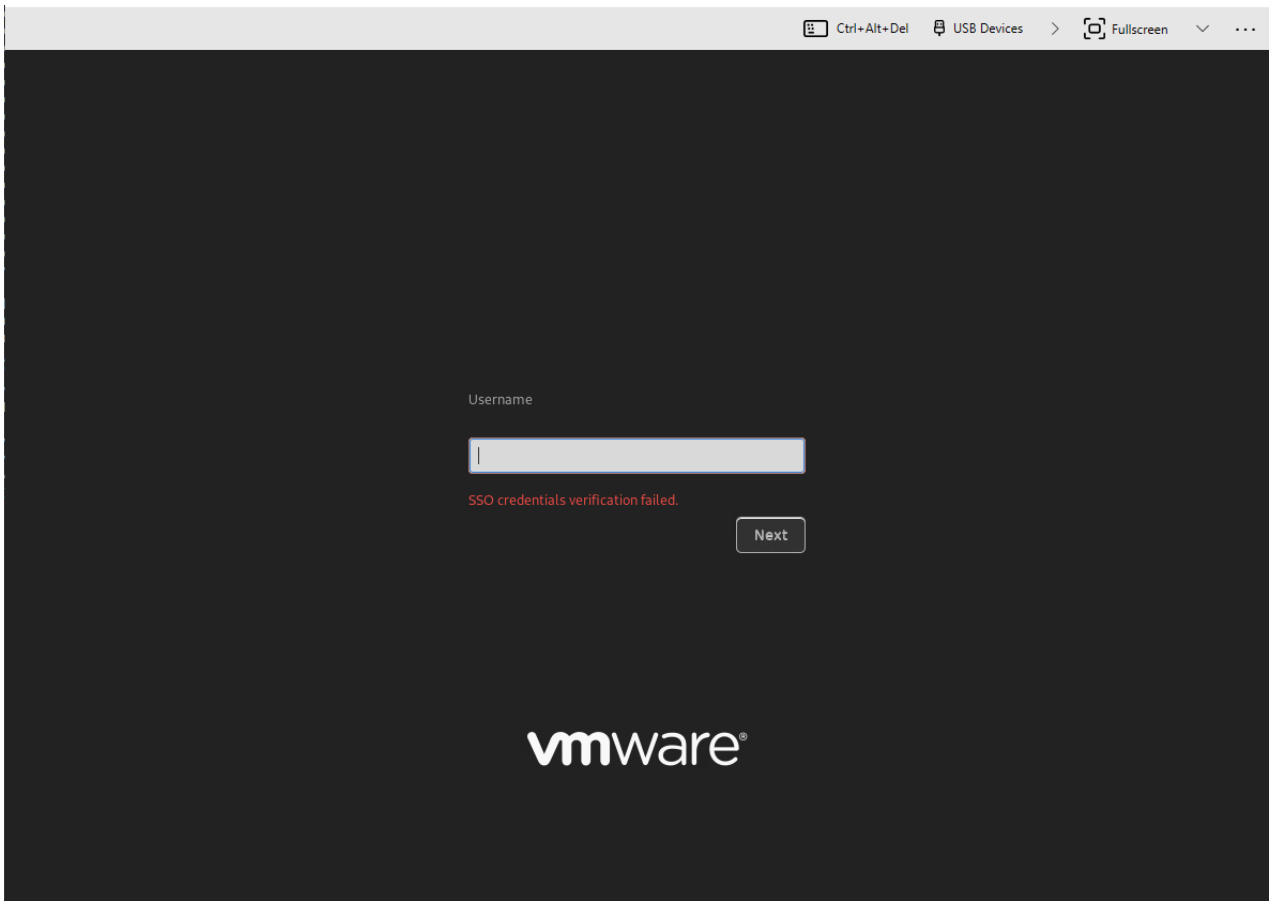
    - RAM: Minimum 2 GB

    - Processor: 2 cores

3. Ensure Network is set to NAT.

4. Save settings.

Step 3: Power On VM:



1. Click Power on this virtual machine.
2. Log into Linux.
3. Open Terminal.

6.Verification:

After login, open Terminal and execute:

- whoami
- pwd
- ls
- uname -a

If commands execute successfully, Linux environment is ready.

7.Basic Configuration:

Update system:

- sudo apt update
- sudo apt upgrade

Install essential tools:

- `sudo apt install build-essential net-tools git curl`

Verify Bash:

- `bash –version`

8. Expected Outcome:

- Linux successfully imported using OVA file.
- Virtual machine configured with proper resources.
- Terminal accessible.
- System ready for Linux command-line and Bash scripting experiments.

9. Result:

The Linux operating system was successfully installed and configured in a virtual environment using VirtualBox / VMware Workstation. The system is functional and ready for performing command-line operations and Bash scripting tasks.